Universitat Rovira i Virgili

Report

Complex Networks

Structural Descriptors of Complex Networks

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https://github.com/sarmijavier/complex_networks_assigments

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1 Introduction

The objective of this assignment is to characterize the macroscopic structure of five distinct networks (net1, net2, net3, net4, net5) by analyzing key descriptors such as degree distribution, clustering coefficient, and path length. This analysis aims to identify relationships between the observed properties and established complex network models (Erdős–Rényi, Watts–Strogatz, Barabási–Albert, and Configuration Model).
In addition, the assignment examines the microscopic structure of the networks by comparing node centrality measures to identify the most influential components. Finally, the study concludes with a spatial analysis of a specific network ($net5$), using its connectivity patterns and visualization to propose a generative algorithm based on random spatial distribution.

2 Structural Characterization

To characterize the macroscopic structure of the four networks provided ($net1$ through $net4$), we analyzed their basic network descriptors and degree distributions. All networks consist of $N = 5000$ nodes with an average degree $\langle k \rangle \approx 10$. Accordingly, they contain a comparable number of edges (approximately $24{,}250 \pm 750$).

2.1 Macroscopic Network Descriptors

The following table summarizes the key structural indicators obtained for each network:

Table 1: Structural descriptors for networks net1 to net4.

Descriptor	**net1**	**net2**	**net3**	**net4**
Nodes (N)	5000	5000	5000	5000
Edges (E)	25000	24873	23508	24975
Min Degree	6	1	3	5
Max Degree	16	24	732	210
Avg Degree $\langle k \rangle$	10.00	9.95	9.40	9.99
Avg Clustering C	0.4141	0.0021	0.0862	0.0107
Assortativity r	-0.0097	-0.0057	-0.1339	-0.0325
Avg Path Length L	5.12	3.96	3.01	3.49
Diameter D	8	7	5	5

All networks exhibit the small-world property, as indicated by average path lengths and diameters that are much smaller than the total number of nodes.
Overall, net3 and net4 have nodes with higher values, such as 732 and 210, which indicate the presence of hubs. Net1 shows the highest value for average clustering, indicating a higher frequency of triangles between nodes. All the networks show connections between different types of nodes due to negative assortativity, but this behavior is most prominent in net3.

2.2 Degree Distribution Representation

The degree distributions provide insight how the nodes are organized:
Net1: This network exhibits a narrow degree distribution tightly centered around the mean ($\langle k \rangle \approx 10$). The homogeneity of the nodes is best visualized using a linear scale histogram, which highlights the absence of "hubs" or extreme outliers. In this system, the vast majority of

nodes possess a nearly identical number of connections, suggesting a highly regular or lattice-like structure.

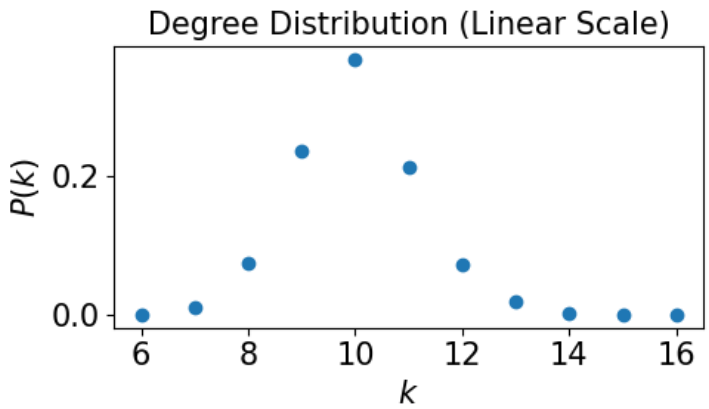


Figure 1: Degree distribution for net1 (Linear Scale).

Net2: Similar to Net1, Net2 demonstrates a high degree of regularity. Although the distribution is slightly broader (spanning from $k \approx 1$ to $k \approx 25$), it remains concentrated around a central value. Utilizing a linear scale allows for a clear observation of the characteristic scale of the network.

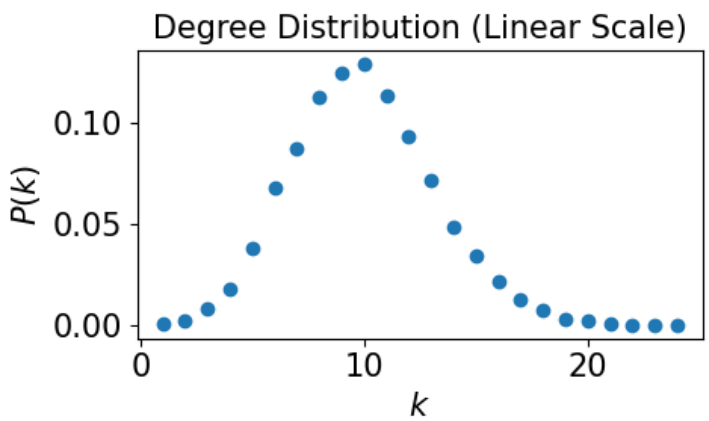
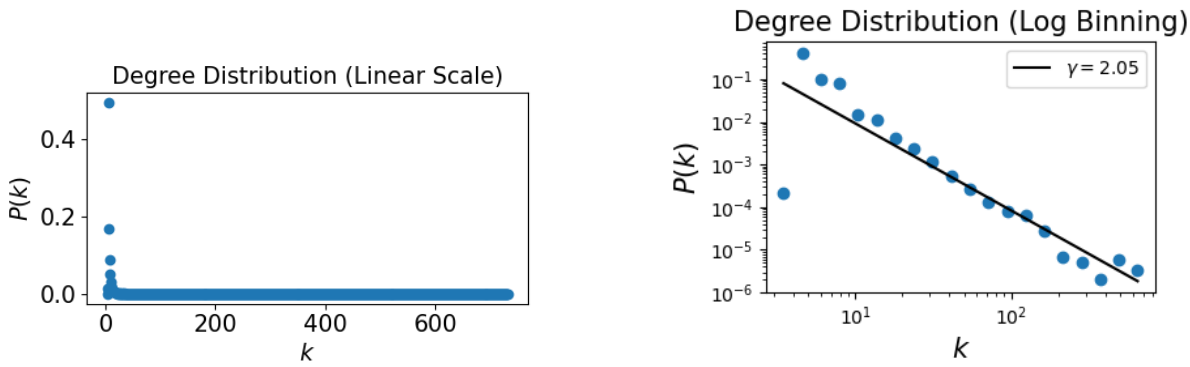


Figure 2: Degree distribution for net2 (Linear Scale).

net3: This network displays a heavy-tailed distribution typical for scale-free networks. In contrast to the homogeneous structures of Net1 and Net2, this topology is characterized by the presence of "hubs". To accurately characterize this behavior, we employed a log-log scale with logarithmic binning. After evaluating several configurations (beginning with 11 bins), we determined that 20 bins provided the most stable results for analysis. This transformation suggests a power-law behavior and estimates the degree exponent γ:"

$$P(k) \sim k^{-\gamma}$$

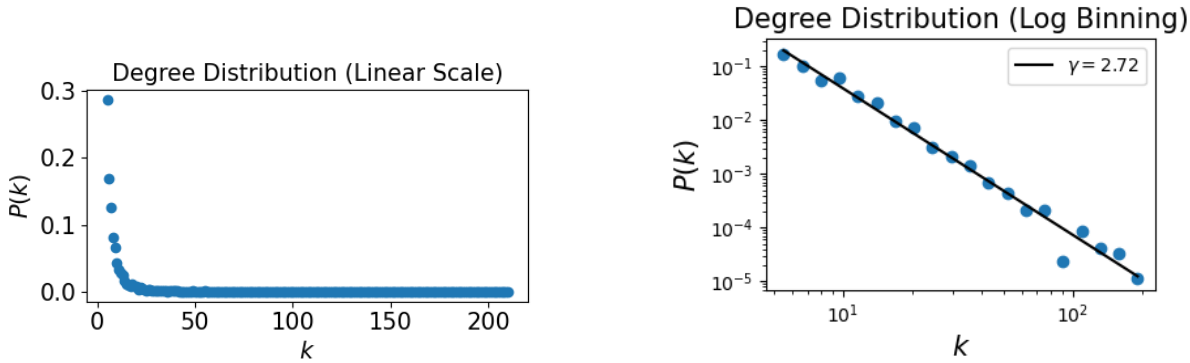


(a) Linear scale distribution.

(b) Log-log distribution.

Figure 3: Degree distributions for Net3: linear and log-log scales.

net4: Similarly, net4 exhibits scale-free properties where a few nodes hold the majority of connections. By plotting the distribution on a log-log scale with 20 bins, the topological heterogeneity of the network becomes evident, allowing for a clear estimation of the degree exponent $\gamma \approx 2.72$.



(a) Linear scale distribution.

(b) Log-log distribution.

Figure 4: Degree distributions for Net4: linear and log-log scales.

2.3 Microscopic Structure: Centrality Analysis

Analyzing the top 5 most central nodes (Degree, Betweenness, and Eigenvector) reveals the following:

Table 2: Top 5 Centrality Values for Network G_{net1}

Metric	Rank	Node ID	Value
Eigenvector	1	651	0.0286
	2	1937	0.0265
	3	4526	0.0256
	4	4398	0.0253
	5	1939	0.0252
Betweenness	1	4747	0.0041
	2	2645	0.0040
	3	230	0.0038
	4	4360	0.0038
	5	1579	0.0037
Degree	1	1693	0.0032
	2	651	0.0030
	3	1579	0.0030
	4	4891	0.0030
	5	41	0.0028

net1: There is significantly less overlap between the centrality rankings compared to heterogeneous networks. While node 651 appears in the top 5 for both Eigenvector and Degree centrality, the leading nodes for Betweenness are entirely distinct. Furthermore, the centrality values across all metrics are relatively low and concentrated in a narrow range. This suggests a uniform distribution of importance, where no single entity dominates the network flow or structure. These results confirm the homogeneity of the topology: having a high local connectivity (Degree Centrality) does not inherently grant a node high global prestige (Eigenvector Centrality) or control over information bottlenecks (Betweenness Centrality). Instead, these structural roles are distributed across different nodes, reflecting a decentralized network.

net2: In contrast to the total decentralization of Net1, Net2 exhibits an emerging structural hierarchy. As shown in Table 3, there is a noticeable overlap among the top-ranked nodes across all three centrality measures.
Node 1581 functions as the primary structural anchor of the network (closely followed by node 787), ranking first in Degree, Betweenness, and Eigenvector centrality. This indicates that 1581 is not only the most popular node (Degree) but also the most critical bridge for information flow (Betweenness) and the most prestigious entity (Eigenvector).
The increased consistency between metrics indicates that Net2 is more centralized than Net1. The fact that a few nodes (such as 1581, 787, and 2375) appear across multiple lists suggests that structural importance is beginning to consolidate. However, because the centrality values remain relatively low and close in magnitude, the network still retains a degree of regularity, stopping short of the extreme "winner-take-all" dynamics found in scale-free systems.

Table 3: Top 5 Centrality Values for Network G_{net2}

Metric	Rank	Node ID	Value
Eigenvector	1	1581	0.0412
	2	3233	0.0343
	3	787	0.0341
	4	2375	0.0337
	5	131	0.0335
Betweenness	1	1581	0.0033
	2	787	0.0026
	3	4382	0.0025
	4	52	0.0023
	5	2375	0.0023
Degree	1	1581	0.0048
	2	787	0.0046
	3	52	0.0042
	4	1990	0.0042
	5	2372	0.0042

Table 4: Top 5 Centrality Values for Network G_{net3}

Metric	Rank	Node ID	Value
Eigenvector	1	5	0.2592
	2	7	0.2417
	3	2	0.2254
	4	0	0.2254
	5	3	0.2070
Betweenness	1	5	0.1378
	2	7	0.1277
	3	0	0.1113
	4	2	0.1093
	5	6	0.0943
Degree	1	5	0.1464
	2	7	0.1382
	3	2	0.1252
	4	0	0.1240
	5	6	0.1110

net3:
As expected, Net3 exhibits the high degree of centralization characteristic of scale-free networks. As shown in Table 4, there is a near-perfect overlap among the top-ranked nodes across all three centrality measures. The same nodes (5, 7, 0, and 2) consistently appear at the top of all three rankings. This indicates that these hubs are not only the most popular entities but also serve as the primary bridges for the entire network and are connected to other highly influential nodes. In contrast with the results for Net1 and Net2, the centrality values in Net3 are significantly higher in magnitude. For instance, the top Eigenvector value reaches approximately 0.26, which is nearly seven times higher than the top value in Net2 (≈ 0.04). This shift illustrates the extreme structural dominance typical of heterogeneous networks, where a few nodes hold a disproportionate amount of influence.

Table 5: Top 5 Centrality Values for Network G_{net4}

Metric	Rank	Node ID	Value
Eigenvector	1	6	0.2239
	2	0	0.2234
	3	9	0.2062
	4	8	0.1741
	5	3	0.1686
Betweenness	1	0	0.0605
	2	6	0.0576
	3	9	0.0553
	4	8	0.0391
	5	3	0.0379
Degree	1	6	0.0420
	2	0	0.0412
	3	9	0.0408
	4	10	0.0326
	5	8	0.0324

net4: Similarly to Net3, Net4 exhibits the high centralization characteristic of scale-free systems. As shown in Table 5, there is a consistent overlap among the top-ranked nodes, with 6, 0, and 9 dominating all three centrality categories. It indicates that all three nodes jointly serve as the network's primary hubs, prestigious centers, and gatekeepers.
Furthermore, the centrality magnitudes in Net4 are lower than those in Net3 (e.g., max betweenness of 0.06 vs. 0.13). This aligns with its larger degree exponent ($\gamma \approx 2.72$), indicating a slightly less extreme concentration of influence compared to the hub dynamics of Net3.

3 Model Identification

Based on the observed descriptors and distributions, we identify the models as follows:

1. **net1: Watts-Strogatz (WS) Model.** Identified by a very high clustering coefficient ($C = 0.41$) and a small diameter relative to N. The degree distribution is very narrow (min 6, max 16), typical of a lattice with partial rewiring.

2. **net2: Erdős–Rényi (ER) Model.** Identified by a very low clustering coefficient ($C \approx 0.002$), which aligns with the theoretical expectation for random graphs where $C \approx \langle k \rangle / N$. The Poisson-like distribution and small average path length confirm a purely stochastic connection process.

3. **net3: Configuration Model (CM).** This network follows a power-law distribution with $\gamma = 2.05 < 2.5$. Within the scope of the models considered in this course, $\gamma < 2.5$ is a distinctive marker for the Configuration Model. Furthermore, the lower edge count, presumably, resulting from the exclusion of self-loops and multi-edges during the construction process supports this classification.

4. **net4: Barabási-Albert (BA) Model.** This network shows a power-law distribution with $\gamma = 2.72$, which is close to the theoretical value of $\gamma = 3$ for the BA model. Nodes with lower IDs serve as primary hubs, indicating they joined the network early (clique start) and accumulated connections over time.

The identification is quantitatively validated by two key scaling laws:

(a) The empirical path length ($L = 3.49$) aligns with the theoretical prediction for $2 < \gamma < 3$: $\langle l \rangle \approx \frac{\ln(5000)}{\ln(\ln(5000))} \approx 3.98$.

(b) The low clustering coefficient ($C = 0.0107$) follows the expected BA scaling $C \sim \frac{(\ln 5000)^2}{5000} \approx 0.0145$, which is significantly lower than values seen in Net3, for example.

4 Analysis of net5

Table 6: Structural descriptors for network net5.

Descriptor	**net5**
Nodes (N)	200
Edges (E)	465
Min Degree	1
Max Degree	10
Avg Degree $\langle k \rangle$	4.65
Avg Clustering C	0.6107
Assortativity r	0.5502
Avg Path Length L	13.13*
Diameter D	35*

*Computed on the largest connected component.

The visualization of *net*5 reveals a network with distinct spatial clustering.

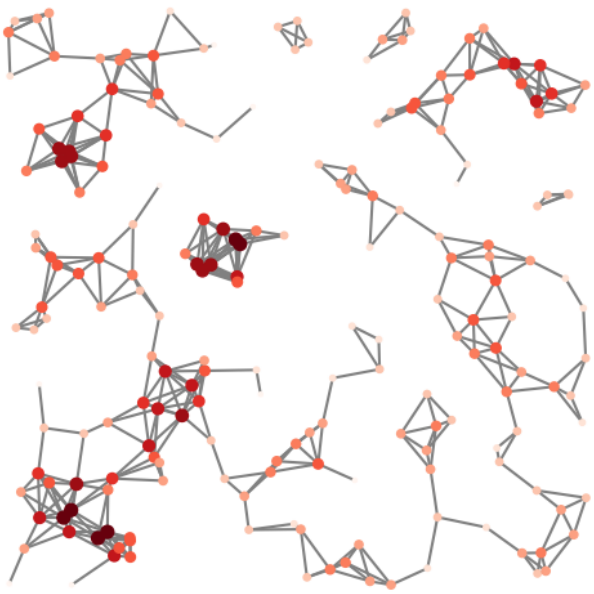


Figure 5: Net 5

- **Connectivity:** The network is not connected, it consists of multiple disjoint components.
- **Scale-Free Property:** The degrees are bounded ($k_{max} = 10$), and the distribution is not heavy-tailed as perfectly seen on the linear scale plot (Figure 6). This network is homogeneous and does not exhibit scale-free properties.
- **Small-World Property:** In the Largest Connected Component, $L = 13.13$ and $D = 35$. These values are very high for a 200-node network. For $N = 200$, a small-world path length would be around $\ln(200) \approx 5.3$.
- **Proposed Algorithm:** Nodes are distributed randomly in a 2D space, and edges are created between any two nodes i, j if their Euclidean distance d_{ij} is less than a chosen radius R. This explains the high clustering $C = 0.61$: if node A is close to B, and B is close to C, there is a very high geometric probability that A is also close to C. This also accounts for the positive assortativity ($r = 0.55$) and the visible spatial clusters. This model is called a Random Geometric Graph.

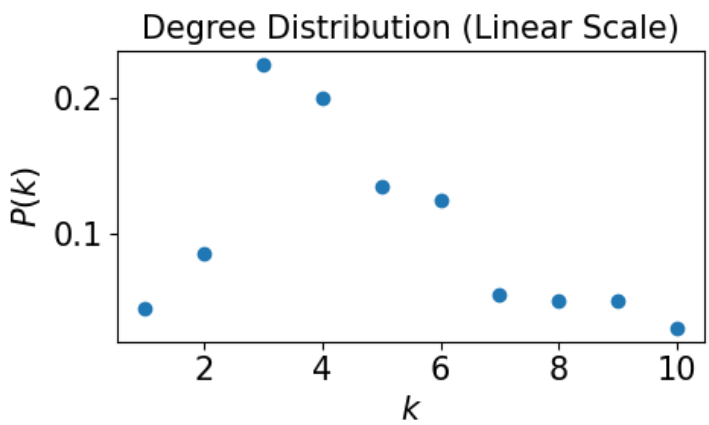


Figure 6: Degree distribution for net5 (Linear Scale).